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A versatile large-scale coherent Ising machine from microwave to visible and telecom wavelength bands

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Abstract

A versatile coherent Ising machine platform based on degenerate optical parametric oscillators has emerged as a powerful approach for solving NP-hard combinatorial optimization problems across diverse applications. Recent work demonstrated breakthroughs using femtosecond laser pumping, achieving a 55% success rate for 100-vertex Max-Cut problems with stable performance maintained for over 8 h. This versatile platform supports fully connected problem mapping and has been successfully validated across multiple domains including molecular docking and credit scoring, substantiating the practical potential of optical approaches for large-scale optimization and paving the way for broad integration into industrial applications.

The Ising model plays a crucial role in statistical physics and computational science¹. Originally developed to describe magnetic ordering, it has evolved into a paradigmatic framework for the study of spin glasses, which are disordered systems characterized by competing interactions and highly rugged energy landscapes. Beyond condensed matter physics, the problem of finding the ground state of an Ising Hamiltonian has wide-ranging applications in computer science, biology, and information processing. Many such problems can be formulated by quadratic unconstrained binary optimization (QUBO), as numerous combinatorial optimization problems can be naturally reformulated as QUBO and then mapped onto equivalent Ising models. From the perspective of computational complexity, the Ising problem belongs to the NP-hard class, posing significant challenges for classical algorithms at large scales.

To address such problems, a variety of heuristic and physics-inspired approaches have been explored. Simulated annealing exploits thermal fluctuations to escape local minima, while quantum annealing and quantum adiabatic computation aim to leverage quantum tunneling effects to improve optimization

performance. Experimental implementations of quantum annealing have been demonstrated in several physical platforms, including superconducting circuits, magnetic systems, and nuclear magnetic resonance setups. While these approaches have achieved notable progress, practical implementations often face constraints in physical connectivity, which can introduce additional overhead when encoding large or densely connected problems².

Motivated by these challenges, alternative physical realizations of Ising machines have been proposed. Among them, optical systems provide a particularly attractive platform. Lasers and degenerate optical parametric oscillator (DOPO) are open, dissipative systems that undergo second-order phase transitions at their oscillation thresholds, naturally supporting collective dynamics and competition among modes. In laser networks, gain competition and cross-saturation favor oscillation modes with minimal threshold gain, a mechanism that can be engineered to reflect the structure of an Ising Hamiltonian. As a result, the system tends to evolve toward low-energy spin configurations³.

DOPOs offer an especially natural representation of Ising spins, as each oscillator above threshold adopts one of two stable phase states⁴. Building on this property, coherent Ising machines (CIMs) based on time-

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multiplexed DOPO have been experimentally demonstrated^{5,6} (Fig. 1). By combining optical coherence with measurement-and-feedback schemes⁷, CIMS enable flexible and scalable implementation of spin–spin interactions⁸, including all-to-all connectivity⁹. This capability allows arbitrary graph-based optimization problems to be mapped directly onto the hardware without the need for complex embedding procedures¹⁰. Owing to their intrinsic parallelism, fast optical dynamics, and high

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